

Knowledge Handoffs: Hearing the Right Answer Is Not the Same as Knowing When It Stops Being Right

Human-readable companion to Knowledge Handoffs Under Change: A Structured Assessment Design for Adaptive Revision

Human-readable companion to “Knowledge Handoffs Under Change: A Structured Assessment Design for Adaptive Revision.” This companion preserves the scientific claims, limits, and contraction rules of the source manuscript while foregrounding the ordinary human question that motivated it. It is not a separate research contribution.

Abstract

A useful answer can be passed from one person to another in seconds even when it took years to learn. That compression is powerful, but it can hide a difference between being able to use an inherited rule today and being able to judge it tomorrow. The Push paper asks what has to survive a knowledge handoff so that a recipient can preserve a rule when superficial details change, restrict it when a case leaves its valid domain, and re-derive an action when the underlying functional relation changes. The proposal does not claim that direct experience is required, that richer explanation is always better, or that a new learning mechanism is needed. Instead, it turns a familiar “I told you” problem into a controlled assessment design. Everyone first reaches the same routine competence. Only then does the handoff vary: rule only, rule plus applicability boundary, or rule plus boundary and rationale. Held-out changes are defined prospectively, and a separate evidence-strength control checks indiscriminate rigidity or revision. The framework survives only if the richer structure reveals selective, held-out differences that simpler transfer models or scores do not already capture.

1. The human problem: “I told you” can be true and still incomplete

A parent, friend, engineer, teacher, or experienced coworker can compress a long learning history into one short warning: “Do this,” “do not do that,” or “this is the part that matters.” The recipient can hear the words, believe the source, remember the answer, and even follow it correctly. Later, after circumstances change, the recipient may still say something like: “I know you told me. I believed you. I just did not understand what you meant until now.”

Push treats that sentence as a measurement problem rather than as proof that the recipient was stubborn or that firsthand experience was necessary. The short answer may have transferred successfully while some of the structure that made the answer appropriately usable did not: why the rule applied, where its boundary was, which changes were harmless, and what kind of change should trigger reconsideration.

correct routine use != adaptive use under change

The paper therefore asks a deliberately practical question: when knowledge is compressed for someone else, what must remain available so that the person can benefit from the inherited answer without becoming unable to recognize when that answer no longer fits?

2. The paper is not arguing that people have to “learn the hard way”

The motivating intuition can easily be misread as a romantic claim that advice never works until a person suffers through the underlying experience. The manuscript explicitly rejects that interpretation. Human learning depends on testimony, instruction, examples, assistance, and inherited knowledge. The point of a handoff is precisely to let one person benefit from what another already learned.

- The source does not have to transmit its complete personal history.
- The recipient does not have to reproduce the source’s emotion, confidence, or subjective experience.
- Assistance and protection are not treated as enemies of learning.
- More information is not assumed to be better in every case.
- A richer handoff earns value only if the later task actually needs the additional structure.

This last point is central. There is no universal knowledge packet that is sufficient for every future demand. A handoff can be completely adequate for today’s routine and inadequate for tomorrow’s exception.

3. “Enough” only makes sense relative to what the recipient will later have to do

Handoff sufficiency is demand-relative.

The manuscript uses handoff sufficiency as a functional description, not as a new mental mechanism. A handoff is sufficient for a particular future demand when enough relevant structure survives for the recipient to use the inherited knowledge appropriately and revise it when the relevant conditions change.

source history -> compressed handoff -> recipient competence -> familiar use -> changed conditions

The familiar-use stage is where an important difference can disappear from observation. Two recipients may be equally accurate, equally fast, equally confident, and equally compliant while the original rule continues to work. Present competence therefore cannot tell us whether both are equally prepared for a later change. The experiment has to create that later change on purpose.

4. Three different kinds of change make the hidden difference observable

Push does not use “change” as one undifferentiated challenge. The experiment declares three structural changes before outcomes are observed. Each asks for a different response to the same inherited rule.

Change	What actually changes	Appropriate response	What the probe asks
Surface / irrelevant change	Appearance or context changes while the rule’s applicability boundary and supporting relation stay intact.	Keep the rule.	Can the recipient avoid overreacting to a difference that does not matter?
Boundary change	The case moves outside the rule’s declared domain while the rule remains valid inside that domain.	Classify the case as out of domain and withhold the rule.	Can the recipient recognize that the inherited answer no longer applies here?
Supporting-relation parameter change	The case remains in domain, but a held-out parameter in the functional relation supporting the old action changes.	Use the retained relation to derive a modified action.	Can the recipient do more than retrieve a taught lookup or exception?

Preserve, restrict, and modify are scoring demands created by the task generator. They are not proposed as three natural kinds of mind. Their purpose is to stop a single “transfer score” from hiding whether a recipient preserved the right structure for a particular kind of future change.

5. Everyone must first be equally competent at the familiar job

The design does not compare novices who received different instruction. Before treatment, every participant receives the same rule-only instruction and the same fixed practice dose under the original regime. Routine competence is assessed against a preregistered criterion. Only after that common baseline are eligible participants randomized to different handoffs.

Handoff	What is transmitted after common competence	What it is intended to make testable
Rule only	The action rule.	Whether routine competence alone is enough for later change.
Rule + boundary	The action rule plus applicability information.	Whether explicit domain limits selectively help with out-of-domain cases.
Rule + boundary + rationale	The rule and boundary plus the general causal or explanatory relation supporting the action.	Whether a retained functional model helps derive an untaught action when an in-domain parameter changes.

Exposure duration, presentation format, and rehearsal opportunity are matched as closely as possible. After randomization, the critical battery provides no corrective feedback or answer-revealing instruction. Otherwise the assessment itself could teach the missing structure and destroy the comparison.

6. The Aster example shows what “re-derive” means

The paper uses a fictional Aster filtration task so every relevant fact can be controlled instead of relying on outside expertise. During the common baseline, an in-domain unit uses cartridge X. The domain is defined as an aqueous feed at or below 8 L/min. The fictional rationale says capture demand is $D = q \times k$, and the appropriate action is the lowest-capacity cartridge whose capacity is at least D.

At baseline, $q = 5$ and $k = 1$, so $D = 5$ and X is adequate. A surface probe changes casing color, display orientation, or other irrelevant features: X should be preserved. A boundary probe increases flow beyond the declared domain: X should be withheld because the rule no longer applies. A relation-parameter probe keeps the unit in domain but changes k to a held-out value. The participant has to use the general relation to calculate a new demand and choose the appropriate cartridge; the critical parameter-action pair was never taught.

The Aster quantities are fictional task-generating facts. They are not claims about real filtration chemistry. The example exists to make lookup, boundary recognition, and genuine use of a transmitted functional relation experimentally separable.

7. Good adaptation includes knowing when not to revise

A person who abandons a valid rule whenever anything surprising happens is not more adaptive than a person who never revises. Push therefore keeps evidence strength separate from the three handoff probes. All groups learn the same evidence-state rule before randomization and later receive weak and strong evidence trials that should respectively favor preservation and provisional revision.

Because every handoff group receives the same evidence-strength mapping, the design predicts no special handoff advantage on this control. Good performance weighs against the simplest “always preserve” or “always revise” strategy. Poor control performance is not interpreted as a unique diagnosis of rigidity, numeracy, attention, comprehension, or any other hidden cause.

8. What a successful pattern would mean - and what it would not mean

A useful result would be selective. Boundary information should matter most when the case leaves the declared domain. Rationale should matter, if at all, when the case remains in domain but requires an action to be derived from a changed functional relation. Surface changes should not reward richer handoffs merely for containing more words.

Even that result would not establish a new learning mechanism. Transfer-appropriate processing, conceptual knowledge, conditional knowledge, preparation-for-future-learning, Cognitive Flexibility Theory, and ordinary content-demand matching are all legitimate conventional explanations. The proposed contribution is the controlled composition: establish equal routine competence first, vary what survives the post-competence handoff, and test held-out demands that require preservation, restriction, or re-derivation.

9. The design has explicit ways to fail and shrink

- If richer handoffs help everything broadly, generic information, attention, memory, or comprehension is the better explanation.
- If a conventional content-by-demand transfer model captures the selective effects, no new statistical architecture is needed.
- If effects disappear on genuinely held-out structural cases, participants may only be following taught instructions or lookups.
- If rule-only participants perform as well as richer groups, the extra transmitted structure was unnecessary for that task.
- If boundary information does not improve prospective out-of-domain recognition and rule withholding, a distinct boundary role has not earned survival.
- If rationale does not outperform boundary information on held-out relation-parameter cases whose action pair was never taught, the proposed generative rationale role has not survived.
- If a finite lookup reproduces the third probe, it contracts to ordinary conditional-policy application.
- If the three probe relations add nothing beyond a simple preserve-versus-revise score, the battery should contract.

These contraction rules matter because the paper is not trying to protect the word “handoff.” The experiment should retain only the distinctions that produce prospective scientific value.

10. The point of Push

Push begins from a social frustration but ends with a methodological rule: do not infer adaptive understanding from routine compliance. A recipient can possess the right answer and still lack some of the structure needed to know when to keep it, where it stops applying, or how to derive a replacement when the relation supporting it changes.

The scientific question is therefore not whether the source explained enough in some absolute sense. It is whether the transmitted structure was sufficient for a declared future demand. The paper succeeds if that framing supports better controlled comparisons than “they knew the rule” or “they did not really understand.” If ordinary transfer science and simpler scoring already do the same work, Push should shrink accordingly.

Selected References

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- The source manuscript contains the complete reference list.*